

MAT 320**Quiz 1****Fall 2025**

1. Factor the polynomial $p(x) = x^3 + 1$ into linear and quadratic factors. The quadratic factor is:
a) $x^2 - x - 1$ b) $x^2 + x - 1$ c) $x^2 + x + 1$ d) $x^2 + 1$ e) $x^2 - x + 1$
Correct Answer: $x^2 - x + 1$
2. Same polynomial $p(x)$ as in the previous question. Use the quadratic formula to find the two complex roots of the quadratic factor. One of them is:
a) $\frac{1}{2} - \frac{\sqrt{3}}{2}i$ b) $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$ c) $\frac{1}{4} - \frac{\sqrt{3}}{4}i$ d) $\frac{1}{4} + \frac{\sqrt{3}}{4}i$ e) $1 + \sqrt{3}i$
Correct Answer: $\frac{1}{2} - \frac{\sqrt{3}}{2}i$
3. Let $q(x) = (x - (a + bi))(x - (a - bi))$ be a quadratic polynomial in factored form. Multiply out so that $q(x) = x^2 + c_1x + c_0$. What is c_1 ?
a) $-2a$ b) $-2bi$ c) $a + bi$ d) $2a + 2bi$ e) $a^2 + b^2$
Correct Answer: $-2a$
4. Same polynomial $q(x)$ as in the previous question. What is c_0 ?
a) $-2a$ b) $-2bi$ c) $a + bi$ d) $2a + 2bi$ e) $a^2 + b^2$
Correct Answer: $a^2 + b^2$
5. Let $f(t) = e^{i4\pi t}$ be a phasor defined for all real numbers t , where t represents time in seconds. What is the frequency of this phasor measured in Hz (cycles per second).
a) 2 b) 2π c) 4 d) 4π e) $\frac{1}{2}$
Correct Answer: 2
6. Same $f(t)$ as in the previous question. What is the (smallest positive) period of f ?
a) 2 b) 2π c) 4 d) 4π e) $\frac{1}{2}$
Correct Answer: $\frac{1}{2}$
7. Let $f_c(z)$ be a complex function $f : \mathbb{C} \rightarrow \mathbb{C}$ which multiplies the input complex variable z by the constant complex number $c = a + bi$. Assume f rotates z by an angle 120 degrees, or $\frac{2}{3}\pi$ radians, counterclockwise, and use this to solve for c . What is the real part a ?
a) 2 b) $-\frac{1}{2}$ c) $\frac{1}{2}$ d) 1 e) -1
Correct Answer: $-\frac{1}{2}$
8. Same $f_c(z)$ as in the previous question. Find the Cartesian form for $f(e^{i\pi/3})$.
a) $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$ b) $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$ c) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ d) 1 e) -1
Correct Answer: -1

9. Let $g(z) = \frac{1}{|z|}$ so that $g : \mathbb{C} \rightarrow \mathbb{R}$. Find $g(1+i)$.

- a) $\frac{\sqrt{2}}{2}$ b) $\frac{3}{2}$ c) $\frac{1}{2}$ d) $\sqrt{2}$ e) 1

Correct Answer: $\frac{\sqrt{2}}{2}$

10. Same $g(z)$ as in the previous question. Find the maximum value of $g(z)$ for inputs $z = x + yi$ where $y = x - 1$.

- a) $\frac{\sqrt{2}}{2}$ b) $\frac{3}{2}$ c) $\frac{1}{2}$ d) $\sqrt{2}$ e) 1

Correct Answer: $\sqrt{2}$